

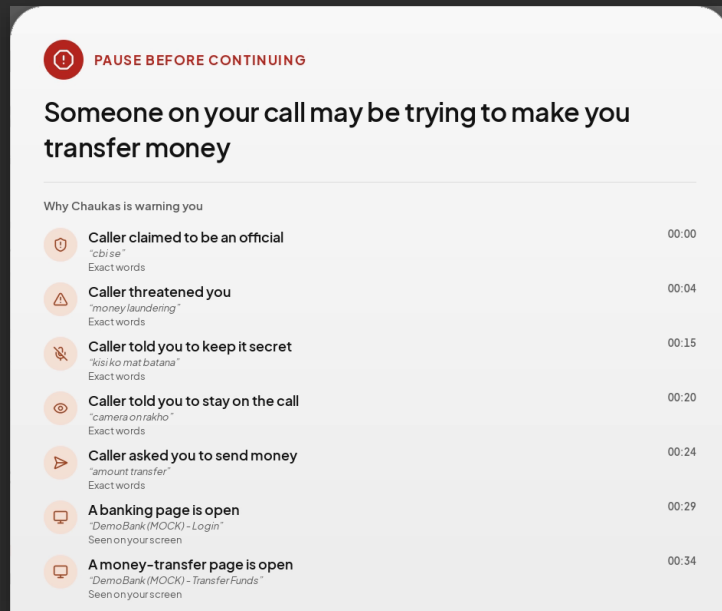
Chaukas

चौकस · alert, watchful

An on-device AI guardian for Windows PCs that notices when a caller is manipulating you, and interrupts before the money, the OTP or your computer is gone.

Devansh Bagaria

github.com/devansh7439/Chaukas · Apache-2.0



Scams hack the decision, not the computer



1. Authority

"This is the CBI cyber cell."



2. Fear

"Your Aadhaar is linked to money laundering."



3. Isolation

"Don't tell your family. Stay on camera."



4. Action

Send money · install AnyDesk · read out the OTP

The victim does every harmful step themselves. Antivirus has nothing to catch: every action is one the user chose.

Existing tools each see one piece



Google Scam Detection

Scam patterns in phone calls, on the phone

Phones only; no link to what happens on screen



Android screen-share warning

Warns when an unknown caller shares the screen

Action only: never hears the conversation



Edge scareware blocker

Blocks full-screen tech-support scam pages

Pages only: a scam that is all talk plus AnyDesk shows none



Chaukas: the conversation and the screen together, on the PC, in English, Hindi and Hinglish, with nothing leaving the device.

Flag suspicious combinations, not suspicious words

“CBI” + “arrest”

could be a news report

Quiet

AnyDesk starts

could be help you asked for

Quiet

CBI officer + “don’t tell anyone” + AnyDesk

social engineering

Alert

$$R = P \times G \times A \times S$$

P pressure: tactics heard, fading over time

G addressed to you, not a news clip

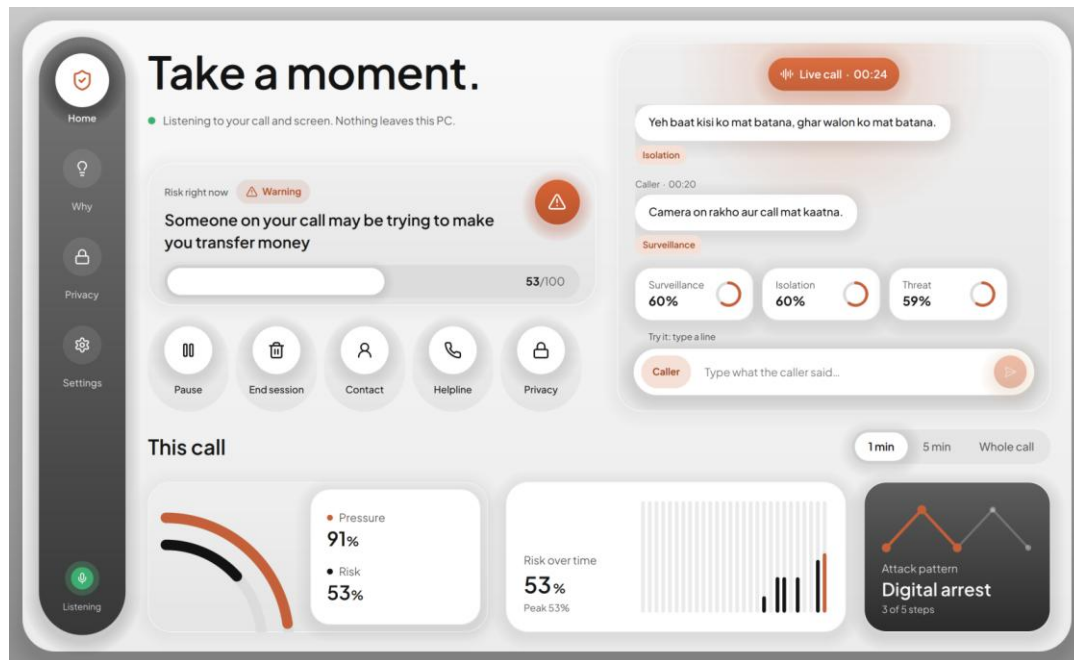
A a matching action on screen

S the attack’s usual order

Every gate ≤ 1 : context can only lower risk.

Three attack chains tracked in parallel: digital arrest, remote access, credential theft. A deterministic engine decides; no generative model can overrule it, and every alert carries its decision trace.

Quiet until it matters, then clear



Notice

A quiet corner card: someone may be pressuring you



Warning

A side panel with the evidence, after a coercive tactic



Critical

A full-screen pause when the whole pattern is there

English and हिन्दी; alerts are also spoken aloud.

Explain every reason; the user decides

Explains

Each reason shows when it happened and what was said or seen, so the user can check it.

Never blocks

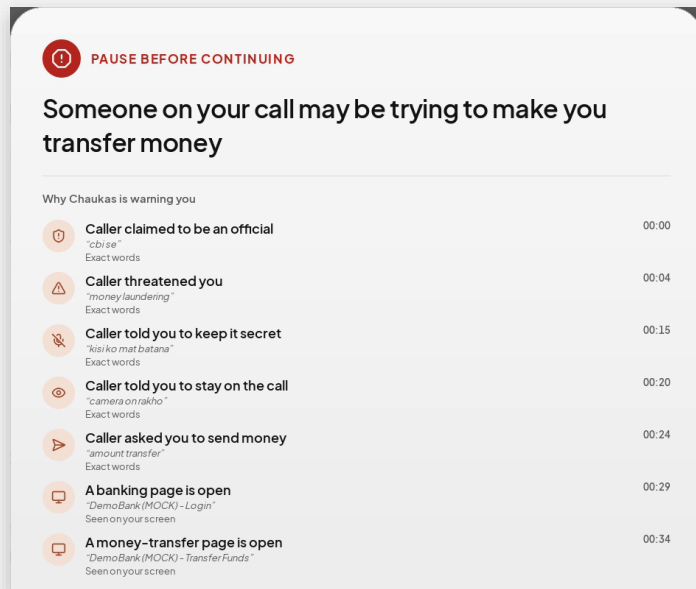
“Continue anyway” unlocks after “I understand this warning”. Chaukas adds friction, not control.

States the fact

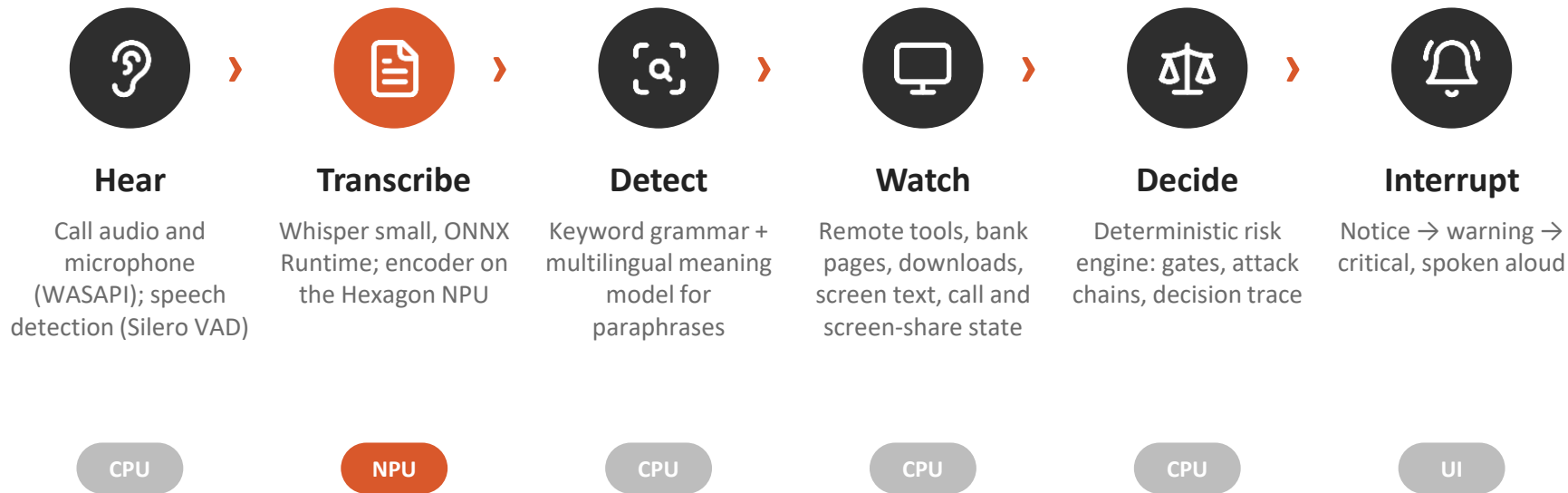
“There is no such thing as a digital arrest.” Safe next steps: hang up, call 1930.

Hides from the scammer

Optional: alerts are excluded from screen sharing, so a remote “agent” can’t see them.



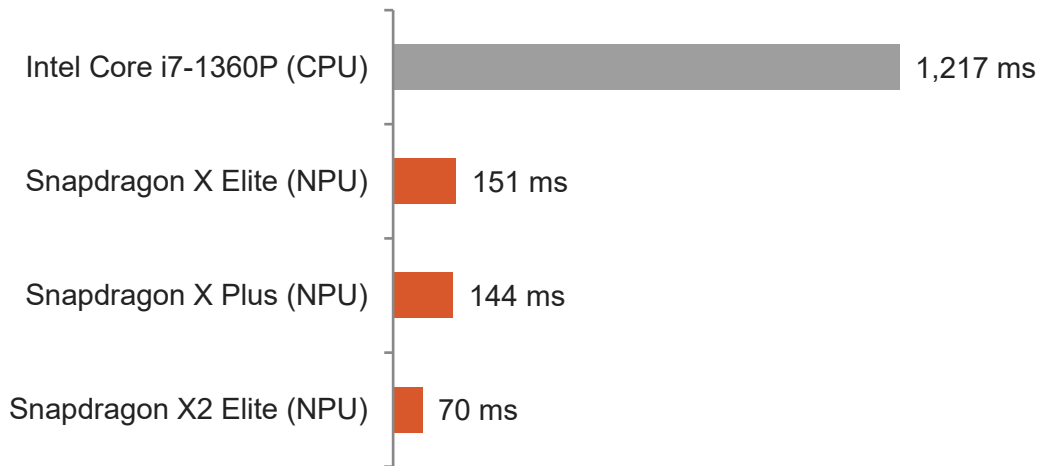
Six steps, all on the PC



Neural work goes to the NPU; the security logic stays deterministic on the CPU. If the NPU can't be used, Chaukas falls back to the CPU and says so: protection never stops.

Whisper's encoder on the Hexagon NPU: 8× faster

Median time per 30 s audio window (lower is better)



313 / 313

encoder layers on the NPU, on every device

24 / 24

spoken test calls got the same alert with the encoder on a real X Elite NPU as on the CPU

Measured on real Snapdragon devices through Qualcomm AI Hub (ONNX Runtime, QNN provider, the same model file Chaukas uses). Every live dependency has a Windows on ARM build. Not yet measured: the whole pipeline on a Snapdragon laptop, and power.

It watches the PC, not just the words



Remote-access tools

AnyDesk, TeamViewer, Quick Assist and more, even when renamed (read from the file's version info).



Bank, transfer and OTP pages

From window titles; the screen text is read only once a call is already suspicious.



Is there a call at all?

A call app holds the microphone; a film doesn't. Films and news with no call: false alarms 7/7 → 0/7.



Screen shared in a browser meeting

“Join the Meet link and share your screen”: scams detected 2/6 → 5/6, no new false alarms.

RESULTS

Measured on fresh test sets, each run once

31 / 36

scams detected across five fresh sets

28 / 33

innocent look-alike calls correctly left quiet
(false alarms: a film, a news report, three
callers asking for a code)

Fresh set	Scams detected	False alarms
Red-team: paraphrase, reordering, adversarial speech	11 / 12	3 / 8
Remote control (tech support, refunds)	6 / 8	0 / 8
Calls vs films and news with no call	6 / 6	0 / 7
Screen shared in a browser meeting	5 / 6	0 / 6
OTP page during a remote session	3 / 4	2 / 4
Spoken aloud, heard by Whisper (English)	11 / 13	2 / 11

Every set was committed to git before its first run. Synthetic scripts, not blind: the same team wrote the detector and the cases. 946 automated tests, 30 security tests, a six-hour soak test, all in CI.

Nothing about a call is saved or sent



Nothing uploaded

Speech-to-text and detection run on the PC. No network use during a call.



Nothing saved

Audio and transcripts stay in memory; old lines are dropped; End session wipes all.



Screen read on alert only

Only once a call is already suspicious; never stored, never logged.



Models checked

Pinned downloads, SHA-256 verified at every load.



Hostile text is harmless

A web page title or the caller's words can't inject into an alert.



Reviewed

Red-team review (8 defects found and fixed), bandit and pip-audit clean.

What Chaukas does not do yet

Calls taken on a phone Chaukas protects calls on the PC.

Remote-control scams with no organisation claim Reach a notice at most.

A delivery agent asking for the order OTP Gets a warning: the same words a scammer uses.

Real Hinglish speech, and the whole pipeline on a Snapdragon laptop Not yet measured (the NPU encoder is).

Test data Synthetic and written by the same team; blind scripts from others are next.

FREE AND OPEN SOURCE TODAY

HP could ship it built in on every Snapdragon PC.



One-click install

install.bat installs, downloads models once, adds a Desktop icon.



A 5-minute demo

Replay a scripted scam call: no microphone, no call needed.



For the people it protects

English and Hindi, spoken alerts, large readable warnings.

Next: the whole pipeline on a Snapdragon laptop, the local LLM on the NPU, blind test scripts, a signed installer.

Chaukas never blocks you. It makes you pause, shows you why, and lets you decide.

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